

FOREST CITY, IA, USA

WMO: 720643

Lat: 43.235N

Lon: 93.624W

Elev: 1229

StdP: 14.05

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 54940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.4	-6.0	-17.3	2.2	-9.9	-13.0	2.8	-5.8	28.8	18.4	26.9	18.0	5.7	330	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	90.2	71.0	87.6	70.6	84.0	69.6	76.6	83.8	74.6	82.3	73.0	80.1	12.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	130.2	80.1	72.4	125.7	78.7	71.5	121.8	77.5	40.9	84.6	39.0	81.8	37.5	80.9	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	22.7	19.8	DB	-20.4	95.4	5.5	3.9	-24.3	98.1	-27.5	100.4	-30.6	102.6	-34.5	105.4	(4)
(5)			WB	-20.8	80.0	5.4	2.1	-24.7	81.5	-27.8	82.7	-30.9	83.9	-34.8	85.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	16.6	19.5	34.0	46.1	58.5	69.0	71.7	68.6	62.6	48.1	34.9	22.9		
	DBStd	21.60	13.14	11.74	12.58	9.63	9.18	6.01	5.75	5.75	7.90	8.78	10.21	11.04		
	HDD50	4061	1036	853	514	183	26	0	0	0	6	143	460	840		
	HDD65	7485	1501	1273	962	569	242	27	12	27	137	526	904	1305		
	CDD50	2665	0	0	18	66	291	569	671	577	383	84	6	0		
	CDD65	616	0	0	1	2	42	146	218	140	64	3	0	0		
	CDH74	5836	0	0	8	30	506	1341	2040	1243	607	61	0	0		
	CDH80	1777	0	0	0	4	179	431	659	330	163	11	0	0		
(14) Wind	WSAvg	8.6	9.8	9.7	10.1	10.7	9.2	7.8	6.1	5.7	6.9	8.6	9.6	8.9		
(15) Precipitation	PrecAvg	30.5	0.7	0.7	1.8	3.0	3.9	4.5	4.3	3.9	3.1	2.1	1.4	0.9		
	PrecMax	41.4	1.9	3.7	3.8	7.1	9.1	12.3	12.3	14.0	9.6	5.0	4.9	2.8		
	PrecMin	16.7	0.0	0.0	0.0	0.8	0.9	0.9	0.6	0.6	0.4	0.1	0.0	0.0		
	PrecStd	6.5	0.6	0.7	1.0	1.5	1.9	2.5	2.6	2.8	2.2	1.3	1.1	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.1	54.8	73.2	78.8	91.3	94.5	93.1	90.4	89.8	81.0	67.9	53.8		
		MCWB	40.0	50.5	58.4	57.8	67.4	71.1	74.9	70.2	68.4	60.0	54.7	47.0		
	2%	DB	37.5	43.0	64.1	72.3	85.6	88.4	90.1	86.5	85.6	73.1	60.6	45.0		
		MCWB	35.3	39.2	53.4	56.7	65.4	68.5	73.2	71.6	69.7	57.5	51.4	41.8		
	5%	DB	36.0	37.2	58.8	68.0	79.3	85.6	86.4	83.6	81.0	68.5	54.4	40.6		
		MCWB	34.3	35.3	50.8	53.9	61.4	67.2	72.8	71.3	67.7	56.9	47.1	37.4		
	10%	DB	33.7	34.4	51.7	63.1	73.4	81.3	82.4	81.2	77.2	63.4	50.0	36.2		
		MCWB	32.8	33.0	46.7	51.6	59.1	66.6	70.8	69.5	65.5	54.1	44.6	34.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.8	50.4	60.5	61.6	71.4	77.7	80.4	77.8	75.1	66.3	57.0	48.0		
		MCDB	45.1	56.9	71.1	71.7	83.6	86.0	87.0	84.9	82.9	70.0	62.3	50.6		
	2%	WB	36.4	40.1	56.4	59.3	68.4	74.4	78.0	75.1	73.1	61.6	52.5	41.5		
		MCDB	38.6	42.8	61.4	68.1	79.1	82.6	84.0	82.7	80.2	68.1	58.9	43.6		
	5%	WB	34.1	35.9	50.7	56.3	65.7	72.2	75.3	73.4	71.2	58.7	47.9	37.2		
		MCDB	35.4	37.9	57.8	65.2	74.3	79.0	82.9	80.4	76.3	65.3	54.0	39.0		
	10%	WB	32.3	33.0	45.8	53.0	63.3	69.7	73.5	71.7	68.9	55.0	44.5	34.6		
		MCDB	33.0	34.1	51.3	60.7	69.8	77.2	80.9	78.3	74.0	62.6	48.9	35.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	16.1	16.2	19.6	21.0	19.9	19.9	20.7	22.8	20.8	17.9	13.7		
		MCDBR	16.2	18.4	28.4	29.4	28.5	25.2	22.1	23.0	26.2	29.7	27.0	18.5		
		MCWBR	13.8	15.3	16.8	16.2	11.8	10.4	10.1	10.7	12.6	15.8	17.4	14.9		
		MCDBR	15.5	17.8	26.4	23.6	21.4	18.8	18.6	18.5	19.5	21.9	25.4	16.0		
	5% WB	MCWBR	13.7	15.0	16.8	14.9	11.0	10.8	10.9	10.3	10.7	15.1	17.8	13.5		
(40) Clear-Sky Solar Irradiance	taub		0.270	0.288	0.311	0.365	0.394	0.421	0.424	0.408	0.376	0.331	0.290	0.268		
	taud		2.425	2.382	2.436	2.322	2.309	2.277	2.247	2.290	2.348	2.469	2.510	2.452		
	Ebn at Noon		278	290	296	285	279	271	268	269	270	271	269	268		
	Edh at Noon		23	29	32	39	40	42	43	39	35	27	22	21		
(44) All-Sky Solar Radiation	RadAvg		577	909	1197	1484	1745	1944	2027	1712	1382	911	606	445		
	RadStd		61	88	119	149	130	151	98	120	95	118	49	31		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page